

PROJECT PROPOSAL FOR NIRAL THIRUVIZHA

1. **Theme :** Med Tech/ Bio Tech/ Health Tech
2. **Problem statement :** How might we develop an app to efficiently maintain and access medico-legal records, reducing search time and manpower required for record management while ensuring easy retrieval and organization of data?
3. **College Code & College Name :**2105-DMI College of Engineering
4. **Guide Name, Designation, Mobile No. & Email id:** Mrs. P. Swarna Lakshmi
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5. **Student Team details:**

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6. **Project Summary:** This project aims to develop a hospital management system with a secured database for medical record storage. The system ensures seamless, encrypted transfer of patient data across hospitals while safeguarding it against unauthorized access and tampering. Leveraging advanced security protocols, such as role-based access control, encryption, and audit logging, the solution ensures compliance with healthcare regulations like HIPAA. The platform's modular design supports scalability and interoperability with existing hospital infrastructure.
7. **Proposed solution with methodology**

This project aims to implement a secure hospital management system to address the challenges of storing and transferring sensitive medical records while maintaining data integrity, privacy, and accessibility. The proposed solution consists of the following methodologies:

 1. **System Architecture**
 - A modular architecture is proposed to separate the storage, access control, and data transfer mechanisms for better scalability and maintainability.

- **Protected Database:** A centralized, encrypted database (e.g., PostgreSQL with AES-256 encryption) is used for secure data storage.
- **API Layer:** RESTful APIs handle all operations, including patient data queries and secure data transfers between hospitals.

2. Security Mechanisms

- **Data Encryption:** All data at rest is encrypted using AES-256, and data in transit is secured with Transport Layer Security (TLS) 1.3.
- **Role-Based Access Control (RBAC):** Provides fine-grained access to ensure only authorized users (e.g., doctors, administrators) can access sensitive data.
- **Audit Trails:** Logs of every data access and modification are maintained to ensure traceability and compliance.

3. Secure Data Transfer

- **Mechanism:** Files and records are securely transferred between hospitals using an encrypted channel.
- **Protocol:** The system uses secure file transfer protocols such as SFTP or HTTPS with additional layers of encryption.
- **Authentication:** Multi-factor authentication (MFA) ensures that only verified hospital systems or personnel can initiate and complete a transfer.

4. Interoperability

- The system integrates with existing Electronic Health Record (EHR) systems using standardized formats like HL7 and FHIR (Fast Healthcare Interoperability Resources) to ensure compatibility.
- A shared schema is used to standardize data representation, reducing the risk of data mismatch or corruption.

5. Compliance with Healthcare Regulations

- The system is designed in compliance with HIPAA, GDPR, and Indian healthcare regulations to ensure data privacy and protection.
- Regular security audits and vulnerability assessments are conducted to validate system security.

6. Implementation Methodology

1. User Access Layer:

- Design and implement a secure authentication mechanism with role-based dashboards for patients and healthcare professionals.

2. Database Security:

- Use database encryption and access policies to ensure data security during storage and retrieval.

3. Secure Transfer Protocols:

- Implement secure APIs and encrypted transfer protocols for data exchange between hospitals.

4. Monitoring and Analytics:

- Develop monitoring tools to analyze access logs and flag suspicious activities for further investigation.

7. Scalability and Future Enhancements

- Implement auto-scaling for cloud-hosted systems to handle large volumes of data during peak periods.
- Design for geo-redundancy to ensure high availability and disaster recovery capabilities.

By incorporating these methodologies, the proposed system ensures secure medical record storage and seamless, encrypted data transfer between hospitals.

8. Workplan / time schedule indicating the project mile stone :

Phase	Description	Period (From – To)
1	Requirement Analysis: Identify key functionalities and data security needs through stakeholder interviews.	01-02-2025 to 07-02-2025
2	Literature Survey: Review existing hospital management systems and security frameworks.	08-02-2025 to 14-02-2025
3	System Design: Create architecture diagrams, database schemas, and data flow designs.	15-02-2025 to 21-02-2025
4	Frontend Development: Develop user interfaces for patient record management and secure data access.	22-02-2025 to 07-03-2025
5	Backend Development: Build API endpoints, implement encryption protocols, and ensure secure storage.	08-03-2025 to 21-03-2025
6	Testing and Deployment: Conduct functional, security, and performance testing, followed by deployment on cloud infrastructure.	22-03-2025 to 31-03-2025

9. Plan of action of implementation

Phase	Description
Requirement Analysis	Collaborate with stakeholders to identify core functionalities and compliance needs.
System Design	Design database schema and define security protocols.
Implementation	Develop and integrate secure storage and data transfer mechanisms.
Testing and Deployment	Conduct thorough testing and deploy the system on secure cloud infrastructure.

10. List of facilities available in the college to develop the prototype of the project

- High-performance computing systems.
- IDEs (Visual Studio Code, IntelliJ IDEA).
- Secure network for development and testing.
- Access to mentors with expertise in healthcare technology and cybersecurity.

11. Nature of Industry support for the project, (if any)

- Partnerships with healthcare technology companies for testing and validation.
- Access to anonymized medical datasets for testing.
- Sponsorship for secure cloud infrastructure.

12. Total Cost

Item	Cost (INR)
Secure Cloud Hosting	5,000
Database Encryption Tools	3,000
Testing Tools and Licenses	2,000
Miscellaneous	2,000
Total	12,000

13. Details of Financial assistance required (Limited to Rs 10,000/-)

Breakdown of Assistance Needed:

- Cloud Hosting: Rs. 5,000/-
- Testing Tools: Rs. 3,000/-
- Miscellaneous: Rs. 2,000/-

14. Expected outcomes / results

- **Impact on Patients:** Enhanced privacy and security of medical records.
- **Impact on Hospitals:** Improved compliance with regulations and reduced administrative burden.
- **Scalability:** Potential to expand to a nationwide hospital network.